

UNIVERSITY OF BAMENDA COLLEGE OF TECHNOLOGY		ACADEMIC YEAR : 2025 – 2026	
		CONTINUOUS ASSESSMENT	
		COURSE MASTER : Eng. MALOBE LOTTIN CYRILLE	
		COURSE TITLE : LINUX PROGRAMMING	
		MONTH :	
		FIELD: CNSM	LEVEL 400
		TIME ALLOWED : 2H	
		INSTRUCTIONS :	
• Answer ALL questions. • Show all necessary steps and commands clearly. • Practical commands must respect Linux syntax. • Write clean and readable shell scripts where required. • Marks are indicated beside each question. • The work should be presented on a manuscript			

SECTION A — LINUX FUNDAMENTALS (6 Marks)

Question 1

a. Define the following terms:

- Linux Kernel
- Shell
- Linux Distribution

(3 Marks)

b. Differentiate between:

- i. CLI and GUI
- ii. Open-source software and proprietary software

(2 Marks)

c. State one major advantage of Linux in:

- Server administration
- Networking environments

(1 Mark)

SECTION B — LINUX COMMAND LINE & FILE SYSTEM MANAGEMENT (8 Marks)

Question 2 — Linux Commands

a. Display the current working directory.

b. List all files including hidden files.

- c. Create a directory named linux_CA.
- d. Move into the directory linux_CA.
- e. Create an empty file called student.txt.
- f. Copy student.txt into /tmp.
- g. Rename student.txt to result.txt.
- h. Remove the file result.txt.

(4 Marks)

Question 3 — File System Navigation

Consider the following Linux path: /home/admin/projects/linux/unit4

Assume you are currently inside: /home/admin

- a. Write the command to move to unit4 using a relative path. (1 Mark)
- b. Write the command to move directly to /etc/network using an absolute path. (1 Mark)
- c. Explain the difference between an absolute path and a relative path. (2 Marks)

SECTION C — FILE PERMISSIONS & PROCESS MANAGEMENT (8 Marks)

Question 4 — Linux File Permissions

Given the following permission: -rwxr-x---

- a. Explain the meaning of each section of this permission. (3 Marks)
- b. Write the command used to:
 - Change file permissions
 - Change file ownership(2 Marks)
- c. Write the command to:
 - Give execute permission to all users on script.sh
 - Remove write permission from group users on script.sh(3 Marks)

Question 5 — Process Management

a. State the function of the following Linux commands:

- ps
- top
- kill

(3 Marks)

b. A process with PID 2450 is consuming excessive CPU resources.

Write the Linux command to terminate the process. (1 Mark)

SECTION D — SHELL SCRIPTING & NETWORK ADMINISTRATION (8 Marks)

Question 6 — Shell Scripting Fundamentals

a. Define Shell Scripting. (1 Mark)

b. State two advantages of shell scripting in Linux system administration. (2 Marks)

Question 7 — Network Diagnostic Shell Script

Write a shell script named `network_check.sh` that performs the following tasks:

- Displays the hostname of the machine
- Displays the IP address configuration
- Tests connectivity to 8.8.8.8
- Tests DNS resolution for google.com
- Displays the routing table or default gateway
- Saves all outputs into a file named `network_report.txt`

Students must write the complete shell script including:

- Shebang line
- Commands
- Output redirection

(5 Marks)

END OF CONTINUOUS ASSESSMENT